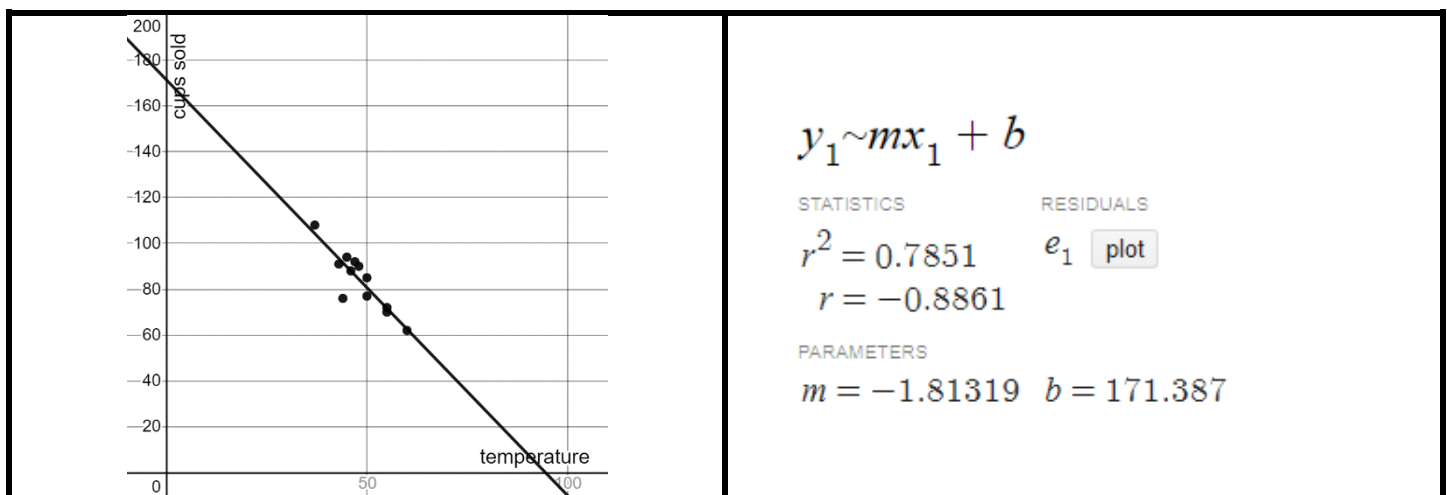


Use this scenario for questions 1 through 6. Polly sells hot chocolate at a stand in a park. She records the number of cups of hot chocolate she sells, as well as the average daily low temperature. The table shown indicates the data for 12 days in January.

Hot Chocolate Sales

Average Temperature (°F)	55	60	55	50	46	37	43	45	47	44	50	48
Cups of Hot Chocolate Sold	70	62	72	85	88	108	91	95	92	76	77	90

Polly used technology to generate a scatter plot and a line of best fit, as shown.



- Write the equation for the line of best fit using the values generated with technology.

$$y = -1.81319x + 171.387$$

- Interpret the slope of the line of best fit in the context.

As the temperature increases, the number of cups of hot chocolate sold decreases.

- Describe the strength of the association between temperature and the number of cups of hot chocolate Polly sells.

Strong

4. Describe the direction of the linear association between temperature and the number of cups of hot chocolate Polly sells.

The direction is negative.

5. Suppose Polly knows that tomorrow's temperature is predicted to be 32°F. Based on the line of best fit, how many cups of hot chocolate should she anticipate selling?

$$y = -1.81319(32) + 171.387$$
$$y = 113 \text{ cups}$$

6. A few weeks ago, Polly sold 104 cups of hot chocolate. Based on the line of best fit, what was the average temperature that day?

$$104 = -1.81319x + 171.387$$
$$x = 37^\circ$$

A scatter plot created with technology generated the following values, as shown. Use the values for questions 7 through 10.

$$y_1 \sim mx_1 + b$$

STATISTICS

$$r^2 = 0.8822$$

$$r = 0.9393$$

PARAMETERS

$$m = 1.45686$$

RESIDUALS

$$e_1$$

$$b = -0.777333$$

7. Describe the strength of the linear correlation for this data.

strong

8. Describe the direction of the linear correlation of this data.

positive

9. Use this data to write the equation of the line of best fit in slope-intercept form.

$$y = 1.45686x - 0.777333$$

10. Use the line of best fit to predict a value for y if the value of x is 9.

$$y = 1.45686(9) - 0.777333$$
$$y = 12.334407$$